UNPACKING THE WEALTH INEQUALITY

Exploring the Effects of Race, Education, & Homeownership

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OVERVIEW OF WEALTH INEQUALITY

Wealth inequality remains a significant issue in the U.S., with disparities across various demographic groups

My research focuses on identifying key drivers that affect wealth inequality. furthermore, how does wealth inequality disproportionately affect various minorities?

Understanding these factors is crucial for addressing systemic disparities and creating policies for economic equity

DATASET

The data set explores wealth inequality of states through a multitude of indicators from the 1960s to 2020

Relevant columns for research:

- wealth_mean
- states
- year
- college
- own
- Black
- Hispanic
- White

aggreagated by researchrs from University of Arizona using surveys from the Federal Reserve

year	state	obs	households	wealth_mea	wealth_med	wealth_top1	wealth_top.1	wealth_top.(	wealth_top.(	wealth_top1	wealth_bott	wealth_gini
1960	alabama	42847	856940	77278.162	35100.75	0.211	0.085	0.034	0.014	0.54	0.062	0.726
1960	alaska	2750	55000	97790.092	36482.414	0.17	0.062	0.026	0.012	0.527	0.244	0.697
1960	arizona	17694	353880	108872.732	45300.973	0.218	0.087	0.036	0.014	0.558	0.069	0.729
1960	arkansas	25150	503000	73961.529	35263.136	0.161	0.059	0.023	0.009	0.507	0.072	0.689
1960	california	237916	4758320	141314.694	60621.91	0.224	0.088	0.036	0.014	0.559	0.058	0.743
1960	colorado	25152	503040	115446.314	54631.22	0.188	0.076	0.031	0.012	0.51	0.074	0.702
1960	connecticut	33892	677840	193005.47	71671.496	0.288	0.117	0.047	0.019	0.619	0.053	0.8
1960	delaware	6212	124240	130939.136	58960.749	0.218	0.082	0.033	0.013	0.547	0.079	0.717
1960	district of co	12115	242300	114314.917	19279.328	0.315	0.128	0.052	0.021	0.681	0.02	0.845
1960	florida	74080	1481600	117124.563	53580.166	0.21	0.084	0.034	0.014	0.534	0.073	0.716
1960	georgia	51696	1033920	84638.569	33943.443	0.228	0.093	0.037	0.015	0.571	0.047	0.756
1960	hawaii	7318	146360	127038.248	27808.356	0.237	0.094	0.038	0.015	0.606	0.025	0.79
1960	idaho	9210	184200	119200.973	64111.222	0.126	0.043	0.017	0.007	0.446	0.095	0.632
1960	illinois	140999	2819980	160848.872	53455.355	0.32	0.131	0.053	0.021	0.631	0.048	0.829
1960	indiana	66737	1334740	107347.713	59074.958	0.173	0.066	0.026	0.011	0.47	0.105	0.648
1960	iowa	40397	807940	114653.949	64441.022	0.15	0.055	0.021	0.009	0.444	0.104	0.629
1960	kansas	32429	648580	111683.128	55830.6	0.168	0.067	0.027	0.011	0.48	0.09	0.667
1960	kentucky	41178	823560	86483.694	39774.483	0.197	0.08	0.032	0.013	0.51	0.075	0.699
1960	louisiana	42248	844960	91446.599	34938.942	0.23	0.094	0.038	0.015	0.576	0.05	0.757
1960	maine	12706	254120	79766.048	39577.568	0.149	0.057	0.022	0.009	0.468	0.251	0.653
1960	maryland	41317	826340	123793.478	56969.323	0.23	0.094	0.038	0.015	0.541	0.076	0.727

vealth_sd	p90	p10	income_mea	house_value	own	rent	own_outrigh	own_mortga	college	white	black	hispanic
234662.739	166974.436	-2126.812	40382.539	48518.843	0.589	0.411	NA	NA	0.139	0.738	0.261	0
187199.257	236162.528	3572.022	66925.968	47580.207	0.456	0.544	NA	NA	0.258	0.83	0.03	0.001
280238.536	236346.705	0	54434.901	64127.141	0.625	0.375	NA	NA	0.247	0.92	0.034	0.003
147355.794	176916.02	-1981.737	32873.483	41309.995	0.602	0.398	NA	NA	0.132	0.812	0.187	0
387565.887	293638.34	-2229.737	63436.406	85868.095	0.565	0.435	NA	NA	0.281	0.931	0.049	0.001
340424.299	259862.158	0	55193.805	70814.371	0.619	0.381	NA	NA	0.271	0.973	0.021	0
530810.916	338062.529	0	69666.291	110371.491	0.577	0.423	NA	NA	0.228	0.963	0.036	0
307720.543	269417.148	0	62859.162	79525.077	0.658	0.342	NA	NA	0.214	0.874	0.123	0
401825.084	231409.722	-3858.237	63214.795	44517.105	0.271	0.729	NA	NA	0.305	0.56	0.433	0
284154.2	247713.024	-1770.46	48362.255	73169.303	0.661	0.339	NA	NA	0.203	0.851	0.147	0
263570.026	186351.053	-2443.561	43967.348	51630.217	0.55	0.45	NA	NA	0.156	0.752	0.248	0
308628.382	295575.147	-3130.354	62985.077	75990.546	0.386	0.614	NA	NA	0.201	0.362	0.007	0.005
179087.288	288946.443	0	46884.485	73348.468	0.693	0.307	NA	NA	0.222	0.988	0.003	0
323583.949	272303.443	-2685.753	62371.222	90481.988	0.54	0.46	NA	NA	0.201	0.907	0.09	0
242982.747	228768.7	2298.387	53665.818	66707.307	0.699	0.301	NA	NA	0.165	0.95	0.049	0
197531.592	255064.723	0	46414.76	60709.518	0.68	0.32	NA	NA	0.178	0.991	0.008	0
245009.651	251481.298	0	49554.231	58537.428	0.679	0.321	NA	NA	0.219	0.962	0.036	0
235231.486	187431.701	0	40690.906	48543.167	0.63	0.37	NA	NA	0.129	0.928	0.072	0
260467.803	199180.645	-2368.763	43088.296	59286.666	0.57	0.43	NA	NA	0.154	0.712	0.286	0
147189.274	182963.131	0	44055.497	49583.391	0.632	0.368	NA	NA	0.152	0.996	0.003	0
363570.136	252573.658	0	63388.685	74367.143	0.629	0.371	NA	NA	0.215	0.853	0.145	0

ESSENTIAL QUESTION

This project looks into three specific indicators:
education, homeownership, and race

Objectives :

- analyze the distribution of wealth across states
- interpret wealth between races
- understand wealth indicators

How does Education, Homeownership, and Race affect the wealth Inequality?

METHODOLOGY

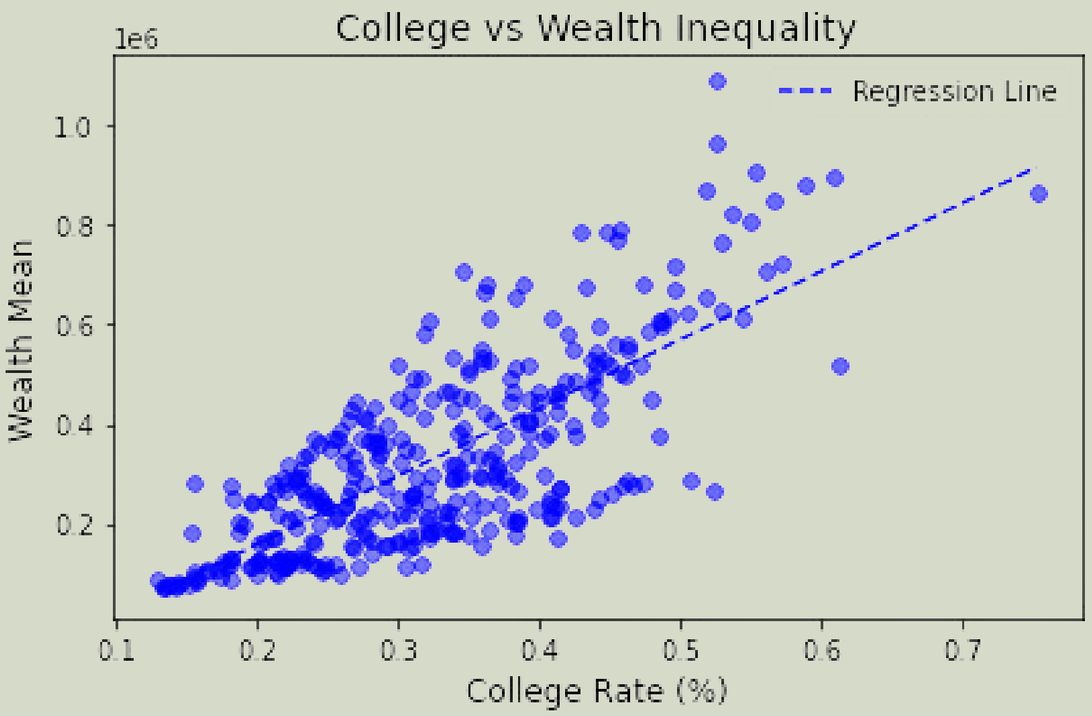
Qualitative Methods

- analyzing categorical data such as Race
- contextualizing data through a written report
- interpreted findings

Quantitative Methods

- statistical analysis
- machine learning models
- linear regresssions

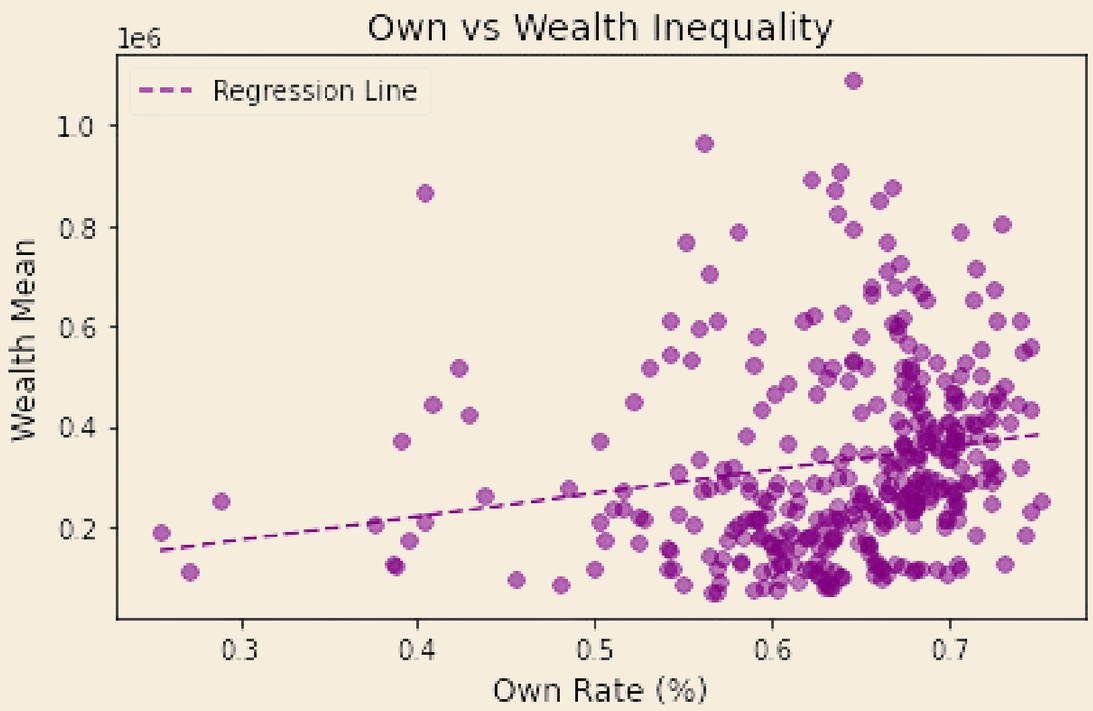
WEALTH VS EDUCATION



a strong indicator of wealth

- strong positive correlation between wealth and education
  - y=−115,016.97+1,369,786.60x
  - r-squared: 0.56

WEALTH VS HOMEOWNERSHIP



a poor indicator of wealth

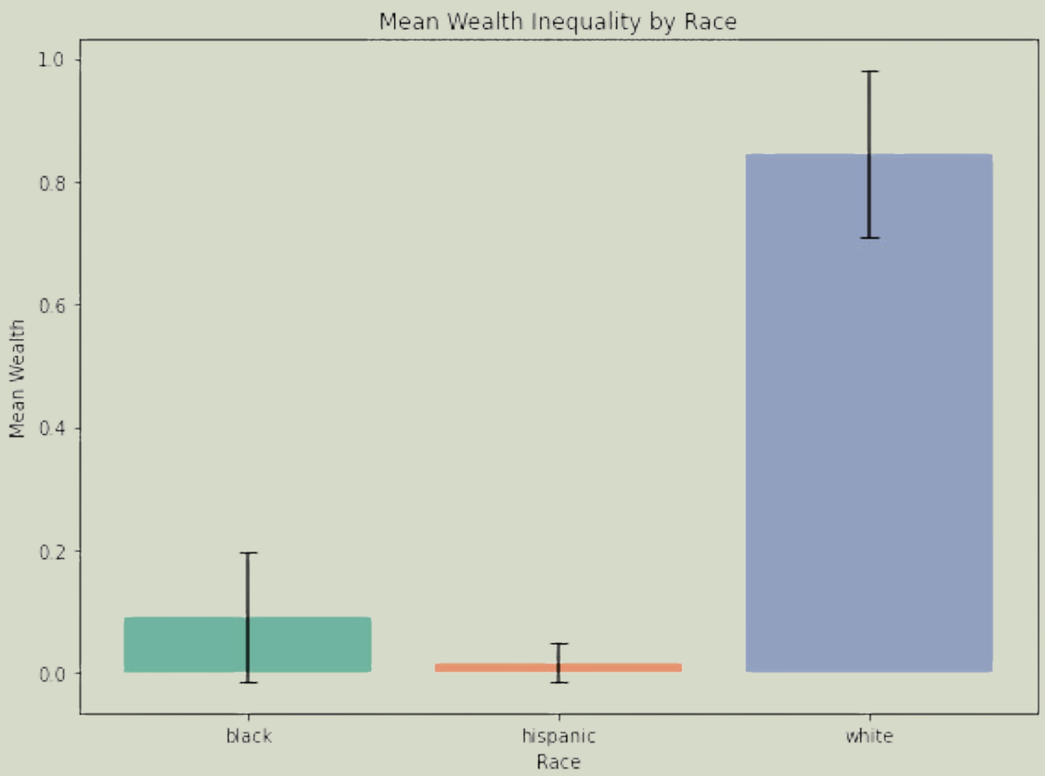
- weak positive correlation between wealth and homeownership
  - y= 34,387.29+465,289.21x
  - r-squared: 0.04

IS RACE A FACTOR?

Through a deep dive into race, we notice that wealth is unfairly distributed across races.

White individuals strikingly have more wealth

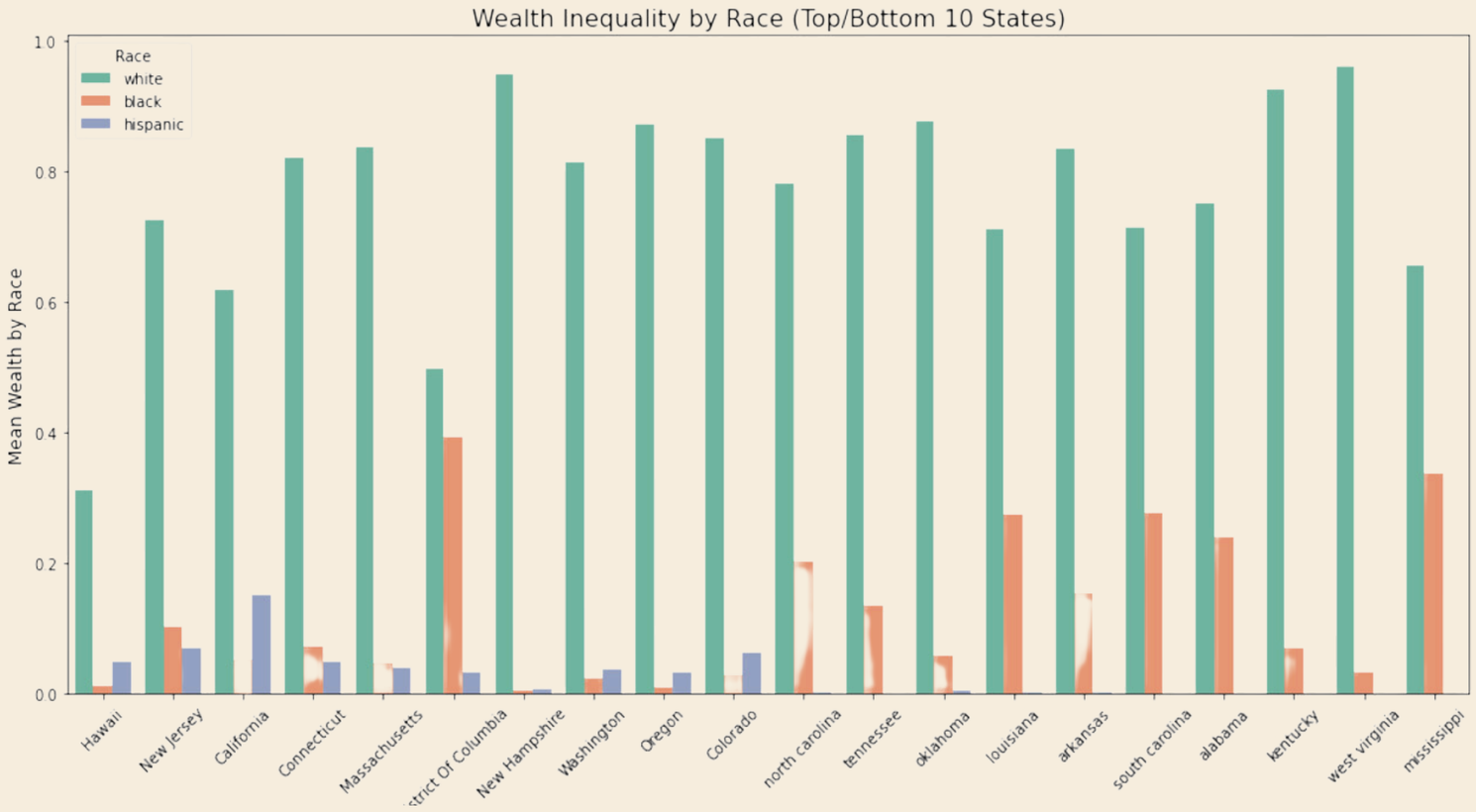
- White - **86.61%** of wealth
- Black - **9.04%** of wealth
- Hispanic - **1.73%** of wealth



WEALTH BY RACE IN STATES

Highest and Lowest States of Wealth

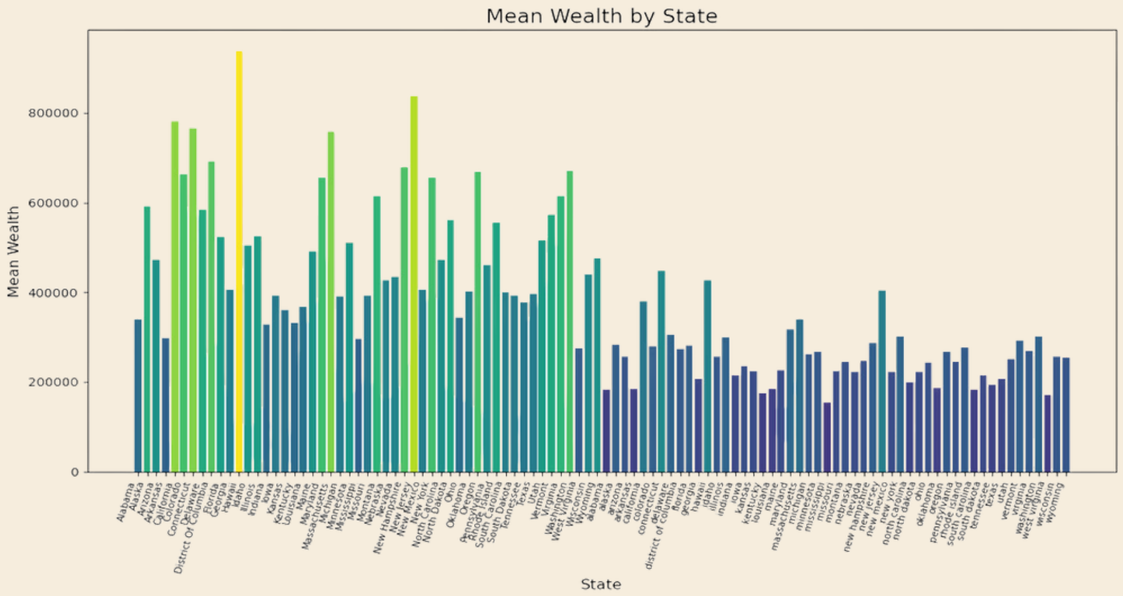
- The insights remain the same in that White individuals hold majority wealth even in states with the least wealth

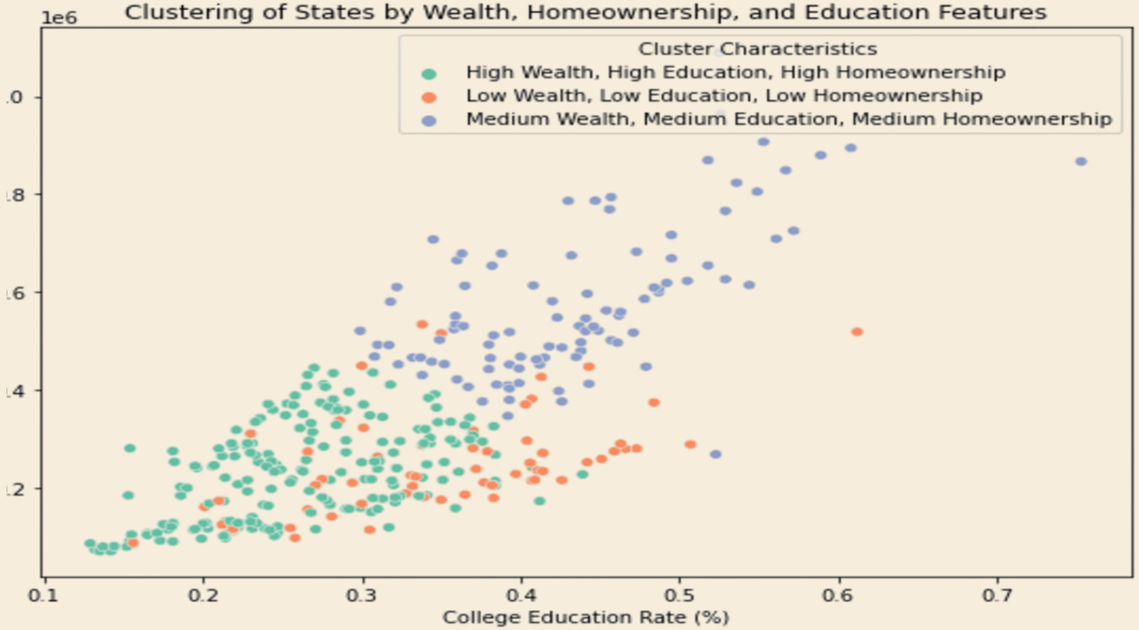


CLUSTER MODEL

The cluster model group states into clusters based on features such as wealth mean, homeownership rate, and college education.

- all high indicators are less dispersed compared to the medium indicators
  - less variation -> consistent values of wealth





Use of KNN classifier and KMeans clustering

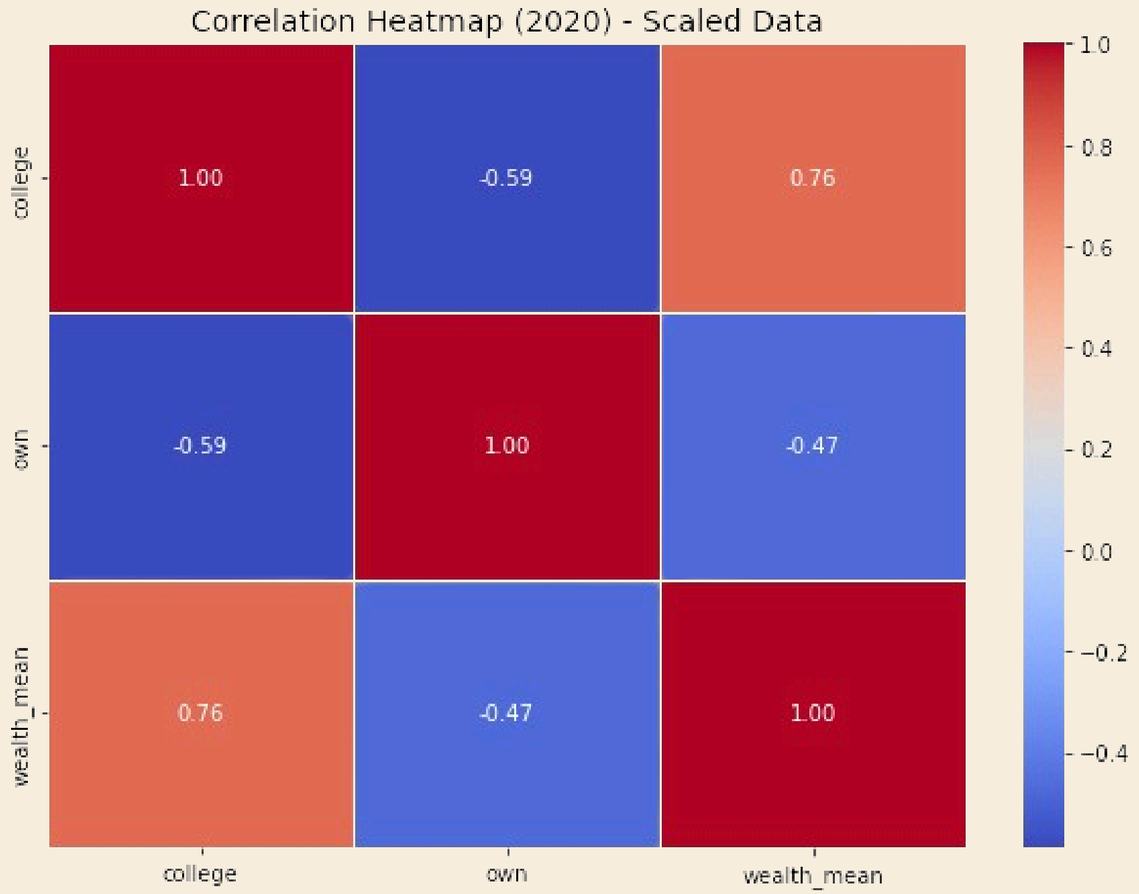
WHAT ABOUT CURRENT DAY?

correlation in 2020:

- *own* -> wealth: -0.47
- *college* -> wealth: 0.76

Insights:

- wealth inequality is still **relevant** today
- education remains to be a **strong indicator** of wealth



KEY FINDINGS

Race is a factor

Wealth disparity is significantly influenced by race, with white people typically having greater wealth than black and Hispanic people

Education is a factor

Wealth disparity and college education are closely associated, and states with greater rates of college education also typically have higher wealth mean values

home-ownership is not a strong factor

Wealth disaprity is weakly associated to homeownership

Relevant to today

Wealt inequality is prevalent today with education still being a strong indicator of wealth.

RECOMMENDATIONS + FUTURE WORK

- **Recommendations**
- look at other contributors of wealth such as the wealth gini
- research underlying reasons for skew in data
- compare results for current year to undertand the impact of current patterns

- **Future Work**
- address systemic barriers
- enhance wealth data especially by race
- collect wealth reports that also monitor policies

THANK YOU

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